

Unit 2 — Oxy-Fuel Pre-Use Safety Checklist

Introduction to Welding Technology (CTE 1042). One copy per outfit, per period. Initial each section. An unticked box means the torch is not lit.

Student: _____ Outfit / bay: _____ Date: _____
Period: _____

1. Before you touch anything

- Bay extraction fan is running.
- Bay is clear of combustibles, including scrap paper and rags.
- Fire extinguisher located and its position stated aloud.
- Cutting goggles (shade 5), safety glasses, gloves, flame-resistant jacket and leather boots on.

2. Cylinders

- Both cylinders chained upright.
- Hydrostatic test dates in date on both cylinders.
- Acetylene cylinder has stood upright at least 30 minutes — it is stored dissolved in acetone and needs to settle.
- Caps fitted on any cylinder in the bay without a regulator.
- Oxygen and fuel gas in storage are separated by 20 ft, or by a 5 ft non-combustible barrier rated for half an hour.

3. Assembly, in this order

- Each cylinder valve cracked briefly to clear the seat.
- Correct regulator on the correct cylinder — acetylene is left-hand thread with a notched nut, oxygen is right-hand.
- Hoses attached: green to oxygen, red to fuel.
- Torch handle fitted, then the cutting attachment, then the tip.
- Both regulators backed out to zero before the cylinder valves are opened.

4. Opening up

- Oxygen cylinder valve opened fully, against the back seat.
- Acetylene cylinder valve opened no more than three quarters of a turn, so it can be shut fast.
- Working pressures set for the tip in use — size 0 tip: 40 psi oxygen, 7 psi acetylene.
- Acetylene is at or below 15 psi. Above 15 psi it becomes unstable. This is not a guideline.

5. Leak test — every joint, every time

- Soap solution applied to both regulator-to-cylinder joints.
- Soap solution applied to both hose-to-regulator and hose-to-torch joints.
- No bubbles anywhere. A bubble means the joint comes apart and is refitted, then retested.

6. Lighting and flame

- Striker in hand. No lighters, no matches, no hot plate.
- Striker held to the side of the tip, never in front of it.
- Flame adjusted to neutral — sharply defined inner cone, no acetylene feather, no harsh hiss.

7. Shutdown, in this order

- Cutting oxygen lever released.
- Torch oxygen valve closed.
- Torch fuel valve closed.
- Both cylinder valves closed.
- Hoses bled by reopening the torch valves briefly.
- Both regulators backed out to zero.
- Hoses coiled and hung, plate returned to the rack, bay swept.

Student initials: _____ Instructor verification: _____

If you notice a backfire (a sharp pop and the flame going out) or a flashback (a squeal with the flame burning back inside the torch), shut the fuel valve first, then the oxygen, and tell the instructor before relighting. Do not relight a torch that has flashed back until it has been checked.